I

(1)

The electric field points from the anode to the cathode, whereas the magnetic field is along the positive z-direction (into the page):

$$\boldsymbol{E} = -\frac{V_p}{d}\hat{\boldsymbol{x}}, \qquad \boldsymbol{B} = B\hat{\boldsymbol{z}}.$$

Using the Lorentz force on an electron, whose charge is $-e$,

$$m\frac{d\boldsymbol{v}}{dt} = -e\left(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B}\right), \qquad \boldsymbol{v} = v_x\hat{\boldsymbol{x}} + v_y\hat{\boldsymbol{y}},$$

and

$$\boldsymbol{v}\times\boldsymbol{B} = Bv_y\hat{\boldsymbol{x}} - Bv_x\hat{\boldsymbol{y}},$$

we obtain

$$\boxed{m\frac{dv_x}{dt} = \frac{eV_p}{d} - eBv_y},$$

$$\boxed{m\frac{dv_y}{dt} = eBv_x}.$$

(2)

Introduce the cyclotron frequency

$$\Omega \equiv \frac{eB}{m}.$$

Then the equations of motion become

$$\frac{dv_x}{dt} = \frac{eV_p}{md} - \Omega v_y, \qquad \frac{dv_y}{dt} = \Omega v_x.$$

Following the complex-variable method, define

$$w(t) \equiv v_x(t) + iv_y(t).$$

The coupled equations can be combined into

$$\frac{dw}{dt} = i\Omega w + \frac{eV_p}{md}.$$

To remove the constant term, define

$$\zeta(t) \equiv i\Omega\left(w(t) - i\frac{V_p}{Bd}\right) = i\Omega w(t) + \frac{eV_p}{md}.$$

It follows that

$$\frac{d\zeta}{dt} = i\Omega\zeta, \qquad \zeta(t) = Ae^{i\Omega t}.$$

The initial conditions $v_x(0) = v_y(0) = 0$ give

$$A = \zeta(0) = \frac{eV_p}{md}.$$

Thus

$$i\Omega(v_x + iv_y) + \frac{eV_p}{md} = \frac{eV_p}{md}\left(\cos\Omega t + i\sin\Omega t\right).$$

Separating the real and imaginary parts yields

$$\boxed{v_x(t) = \frac{V_p}{Bd}\sin\left(\frac{eB}{m}t\right)},$$

$$\boxed{v_y(t) = \frac{V_p}{Bd}\left[1 - \cos\left(\frac{eB}{m}t\right)\right]}.$$

(3)

Using $x(0) = y(0) = 0$, we integrate the velocity components:

$$\begin{aligned} x(t) &= \int_0^t v_x(t')\,dt' \\ &= \frac{V_p}{Bd}\int_0^t \sin(\Omega t')\,dt' \\ &= \boxed{\frac{mV_p}{eB^2d}\left[1 - \cos\left(\frac{eB}{m}t\right)\right]}, \end{aligned}$$

$$\begin{aligned} y(t) &= \int_0^t v_y(t')\,dt' \\ &= \frac{V_p}{Bd}\int_0^t [1 - \cos(\Omega t')]\,dt' \\ &= \boxed{\frac{V_p}{Bd}\left[t - \frac{m}{eB}\sin\left(\frac{eB}{m}t\right)\right]}. \end{aligned}$$

(4)

The maximum displacement along the x-direction is

$$x_{\max} = \frac{mV_p}{eB^2d}\max_t[1 - \cos(\Omega t)] = \frac{2mV_p}{eB^2d}.$$

For the electron to reach the anode at $x = d$, we require

$$x_{\max} \geq d.$$

Therefore,

$$\boxed{V_p \geq \frac{eB^2d^2}{2m}}.$$

Equality corresponds to the electron just reaching the anode at the turning point of its x-motion.

Note. The trajectory is a cycloid-like curve: the x-motion is periodic, while the y-coordinate contains a uniform drift with speed $V_p/(Bd)$ in the positive y-direction.

II

(5)

The continuity equation expresses local charge conservation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \boldsymbol{j} = 0.$$

Since the system is one-dimensional and in a steady state,

$$\frac{\partial \rho}{\partial t} = 0, \qquad \frac{dj_x}{dx} = 0.$$

The conventional current points towards the cathode, so $j_x = -J$. Consequently,

$$\boxed{\frac{dJ}{dx} = 0},$$

i.e., the current density is independent of x.

(6)

Electrons move along the positive x-direction, whereas the chosen positive current direction is along the negative x-direction. Because $\rho(x) < 0$, the current density is

$$\boxed{J = -\rho(x) v(x)}.$$

(7)

By conservation of mechanical energy,

$$\frac{1}{2} m v(x)^2 - eV(x) = \frac{1}{2} m v(0)^2 - eV(0) = 0.$$

Thus,

$$\boxed{\frac{1}{2} m v(x)^2 = eV(x)}, \qquad \boxed{v(x) = \sqrt{\frac{2eV(x)}{m}}}.$$

(8)

The one-dimensional Poisson equation reads

$$\frac{d^2 V}{dx^2} = -\frac{\rho(x)}{\varepsilon}.$$

Using $\rho(x) = -J/v(x)$ and the result of (7),

$$\begin{aligned} \frac{d^2 V}{dx^2} &= \frac{J}{\varepsilon v(x)} \\ &= \frac{J}{\varepsilon} \sqrt{\frac{m}{2eV(x)}}. \end{aligned}$$

Therefore,

$$\boxed{\frac{d^2 V}{dx^2} = \frac{J}{\varepsilon} \sqrt{\frac{m}{2e}}\, V^{-1/2}}.$$

(9)

Multiplying the equation in (8) by dV/dx gives

$$\frac{dV}{dx}\frac{d^2V}{dx^2} = \frac{J}{\varepsilon}\sqrt{\frac{m}{2e}}V^{-1/2}\frac{dV}{dx}.$$

Since

$$\frac{dV}{dx}\frac{d^2V}{dx^2} = \frac{1}{2}\frac{d}{dx}\left(\frac{dV}{dx}\right)^2,$$

we can integrate with respect to x:

$$\frac{1}{2}\left(\frac{dV}{dx}\right)^2 = \frac{J}{\varepsilon}\sqrt{\frac{m}{2e}}\int V^{-1/2}\,dV + C.$$

Here, $\int V^{-1/2}\,dV = 2\sqrt{V}$. The boundary conditions $V(0) = 0$ and $V'(0) = 0$ imply $C = 0$. Hence

$$\frac{1}{2}\left(\frac{dV}{dx}\right)^2 = \frac{2J}{\varepsilon}\sqrt{\frac{m}{2e}}\sqrt{V},$$

or

$$\boxed{\left(\frac{dV}{dx}\right)^2 = \frac{4J}{\varepsilon}\sqrt{\frac{m}{2e}}\sqrt{V}}.$$

As the potential increases from cathode to anode, we take the positive square root:

$$\boxed{\frac{dV}{dx} = 2\sqrt{\frac{J}{\varepsilon}}\left(\frac{m}{2e}\right)^{1/4}V^{1/4}}.$$

Note. The factor 2 from $\int V^{-1/2}\,dV = 2\sqrt{V}$ must be kept; it is responsible for the factor 4 in the squared-derivative equation.

(10)

Separating variables in the final equation of (9),

$$V^{-1/4}\,dV = 2\sqrt{\frac{J}{\varepsilon}}\left(\frac{m}{2e}\right)^{1/4}dx.$$

Integrating from $x = 0$ to $x = d$, where $V(0) = 0$ and $V(d) = V_p$,

$$\frac{4}{3}V_p^{3/4} = 2\sqrt{\frac{J}{\varepsilon}}\left(\frac{m}{2e}\right)^{1/4}d.$$

Squaring and solving for J gives

$$\boxed{J = \frac{4}{9}\varepsilon\sqrt{\frac{2e}{m}}\frac{V_p^{3/2}}{d^2}}.$$

This is the *Child–Langmuir law* for the space-charge-limited current between parallel plates.

Optional check. Integrating the same differential equation to an arbitrary x yields

$$V(x) = V_p\left(\frac{x}{d}\right)^{4/3}.$$

It satisfies $V(0) = 0$, $V(d) = V_p$, and $V'(0) = 0$, providing a useful consistency check on the result.